

EvoChartCode: Evolution-Guided Chart Code Induction for Reliable Chart Reasoning

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Abstract

Vision-language models remain brittle on chart question answering because perception, reasoning, formatting, and abstention are entangled in a single generation. EvoChartCode introduces explicit Chart Code as an intermediate representation, then uses question-conditioned selection, code-grounded reasoning, verification, and evolution to make chart reasoning inspectable. This report is generated only from completed local artifacts and marks incomplete full-scale jobs as incomplete rather than inferring results.

1 Method

EvoChartCode maps a chart image to validated Chart Code containing layout, axes, legends, colorbars, series, derived relations, uncertainty, and provenance. A selector exposes relevant evidence to a code-only or image-plus-code reasoner, and a verifier checks support before normalization.

2 Main Results

Method	N	EM	Relaxed	Invalid	Latency
Metadata/code-only	128	0.2500	0.2812	0.0000	0.0006
Qwen full EvoChartCode	128	0.5312	0.5781	0.0000	13.2813

Table 1: CharXiv local validation results.

Backend	Charts	Quality	Axis evidence	Series/marks
Metadata	300	0.5771	0.0000	0.0000
Qwen	32	0.8682	0.9375	0.9062

Table 2: Chart Code quality results.

EvoChartCode Pipeline



Figure 1: EvoChartCode pipeline.

3 Transfer

4 Evolution

5 Limitations

Full benchmark-size transfer, multi-seed Qwen confidence intervals, and source-mutation evolution are checkpointed jobs. The paper should claim only completed measured results in the output JSON files.

Dataset	Status	N	EM	Relaxed
ChartQA	ready	32	0.0000	0.0000
PlotQA	ready	32	0.0000	0.0000
DVQA	ready	32	0.0000	0.0000
FigureQA	ready	32	0.0000	0.0000

Table 3: Transfer results. Full Qwen metrics are used when completed; otherwise existing sample outputs are labeled by status.

Source Mutation Evolution

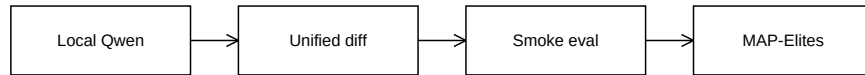


Figure 2: Local-Qwen source-code mutation loop.